

Project: Reconstruction from Non-Uniform Samples Using a DCT- ℓ_p Prior Majorization–Minimization with ADMM (MM–ADMM)

1. Problem

Let $x^* \in \mathbb{R}^N$ be a 256×256 grayscale image (vectorized). We observe only

$$m = Wx^* + \eta \in \mathbb{R}^M, \quad M \ll N,$$

where W selects M pixel locations (non-uniform sampling). The matrix $W^\top W$ is a diagonal 0/1 mask in the *image domain*. We reconstruct x by

$$J(x) = \|Wx - m\|_2^2 + \lambda \sum_{i=1}^N (\varepsilon + |(\text{DCT } x)_i|)^p, \quad p \in (0.4, 1), \varepsilon > 0, \lambda > 0. \quad (1)$$

Here DCT/IDCT are the orthonormal 2D DCT/IDCT (so $\text{IDCT DCT} = \text{DCT IDCT} = I$). The prior is a smoothed, nonconvex, *concave in* $|\cdot|$ sparsity promoter; as $p \rightarrow 1$ it approaches ℓ_1 , for smaller p it promotes stronger sparsity.

2. MM surrogate (IRL1)

Define DCT coefficients $\hat{y} = \text{DCT } x$. For the separable penalty $\phi(|u|) = (\varepsilon + |u|)^p$ with $p \in (0.4, 1)$, concavity in $|u|$ yields the global tangent majorizer

$$\phi(|u|) \leq \phi(|u_0|) + \phi'(|u_0|)(|u| - |u_0|),$$

so at iterate k ,

$$\sum_i (\varepsilon + |\hat{y}_i|)^p \leq \sum_i w_i^{(k)} |\hat{y}_i| + \text{const}, \quad w_i^{(k)} := \phi'(|\hat{y}_i^{(k)}|) = p(\varepsilon + |\hat{y}_i^{(k)}|)^{p-1} > 0.$$

Dropping constants gives the convex surrogate

$$\min_x \underbrace{\|Wx - m\|_2^2}_{\text{data term}} + \lambda \sum_i w_i^{(k)} |(\text{DCT } x)_i|, \quad (2)$$

i.e., a *weighted* ℓ_1 (IRL1) problem for fixed weights $w^{(k)}$.

3. ADMM for the surrogate (image/coefficient split)

Introduce $v = \text{DCT } x$ and enforce $v = \text{DCT } x$. For fixed $w^{(k)}$, solve

$$\min_{x,v} \|Wx - m\|_2^2 + \lambda \sum_i w_i^{(k)} |v_i| \quad \text{s.t.} \quad v = \text{DCT } x.$$

Scaled ADMM with dual u and penalty $\rho > 0$:

$$\mathcal{L}_\rho(x, v, u) = \|Wx - m\|_2^2 + \lambda \sum_i w_i^{(k)} |v_i| + \frac{\rho}{2} \|\text{DCT } x - v + u\|_2^2 - \frac{\rho}{2} \|u\|_2^2.$$

Updates (all matrix-free). Let $M := W^\top W$ be the 0/1 mask image. Using orthonormal DCT:

$$\|\text{DCT } x - v + u\|^2 = \|x - \text{IDCT}(v - u)\|^2.$$

- **x -update (closed form, pointwise in image domain):**

$$x^{t+1} = \arg \min_x \|Wx - m\|_2^2 + \frac{\rho}{2} \|x - s^t\|_2^2, \quad s^t := \text{IDCT}(v^t - u^t).$$

Since M is diagonal with entries $M_j \in \{0, 1\}$,

$$x_j^{t+1} = \begin{cases} \frac{m_j + \rho s_j^t}{1 + \rho}, & M_j = 1 \quad (\text{observed pixel}), \\ s_j^t, & M_j = 0 \quad (\text{unobserved pixel}). \end{cases}$$

- **v -update (weighted soft-threshold in DCT domain):**

$$v^{t+1} = \arg \min_v \lambda \sum_i w_i^{(k)} |v_i| + \frac{\rho}{2} \|v - z^{t+1}\|_2^2, \quad z^{t+1} := \text{DCT } x^{t+1} + u^t,$$

i.e.

$$v_i^{t+1} = \text{soft}(z_i^{t+1}, (\lambda/\rho) w_i^{(k)}) := \text{sign}(z_i^{t+1}) \max(|z_i^{t+1}| - (\lambda/\rho) w_i^{(k)}, 0).$$

- **Dual update:**

$$u^{t+1} = u^t + \text{DCT } x^{t+1} - v^{t+1}.$$

Outer MM loop. After ADMM converges for weights $w^{(k)}$, set $\hat{y}^{(k+1)} := v^{(\text{final})}$ (or $\text{DCT } x^{(\text{final})}$) and update

$$w_i^{(k+1)} = p(\varepsilon + |\hat{y}_i^{(k+1)}|)^{p-1}.$$

Repeat until outer relative change is small.

Algorithm 1 MM-ADMM for $J(x) = \|Wx - m\|_2^2 + \lambda \sum_i (\varepsilon + |(\text{DCT } x)_i|)^p$

Require: m , mask $M = W^\top W$ (0/1 array), DCT/IDCT; parameters $p \in \{0.5, 0.7, 0.9\}$, $\varepsilon = 10^{-6}$, $\rho > 0$, grid of λ values

- 1: $x^{(0)} \leftarrow W^\top m$ ▷ zero-filled image
- 2: $\hat{y}^{(0)} \leftarrow \text{DCT}(x^{(0)})$
- 3: $w_i^{(0)} \leftarrow p (\varepsilon + |\hat{y}_i^{(0)}|)^{p-1}$ for all i
- 4: **repeat** ▷ Outer MM loop
- 5: $x^0 \leftarrow x^{(k)}$, $v^0 \leftarrow \hat{y}^{(k)}$, $u^0 \leftarrow 0$
- 6: **repeat** ▷ Inner ADMM loop
- 7: $s^t \leftarrow \text{IDCT}(v^t - u^t)$
- 8: **(x-update)** $x_j^{t+1} \leftarrow \begin{cases} \frac{m_j + \rho s_j^t}{1 + \rho}, & M_j = 1 \\ s_j^t, & M_j = 0 \end{cases}$ for all pixels j
- 9: $z^{t+1} \leftarrow \text{DCT}(x^{t+1}) + u^t$
- 10: **(v-update)** $v_i^{t+1} \leftarrow \text{soft}(z_i^{t+1}, (\lambda/\rho) w_i^{(k)}) = \text{sign}(z_i^{t+1}) \max(|z_i^{t+1}| - (\lambda/\rho) w_i^{(k)}, 0)$
- 11: **(dual)** $u^{t+1} \leftarrow u^t + \text{DCT}(x^{t+1}) - v^{t+1}$
- 12: **until** ADMM primal/dual residuals $\leq 10^{-4}$ **or** max iters reached
- 13: $x^{(k+1)} \leftarrow x^{t+1}$, $\hat{y}^{(k+1)} \leftarrow v^{t+1}$
- 14: $w_i^{(k+1)} \leftarrow p (\varepsilon + |\hat{y}_i^{(k+1)}|)^{p-1}$ for all i
- 15: **until** $\frac{\|x^{(k+1)} - x^{(k)}\|}{\max(1, \|x^{(k)}\|)} \leq 10^{-3}$ **or** max outer iters reached
- 16: **Output:** $\hat{x} = x^{(k+1)}$

5. Experimental setup and deliverable

This part should be parallel with corresponding parts of the first assignment. But here the value of p should be doubled.